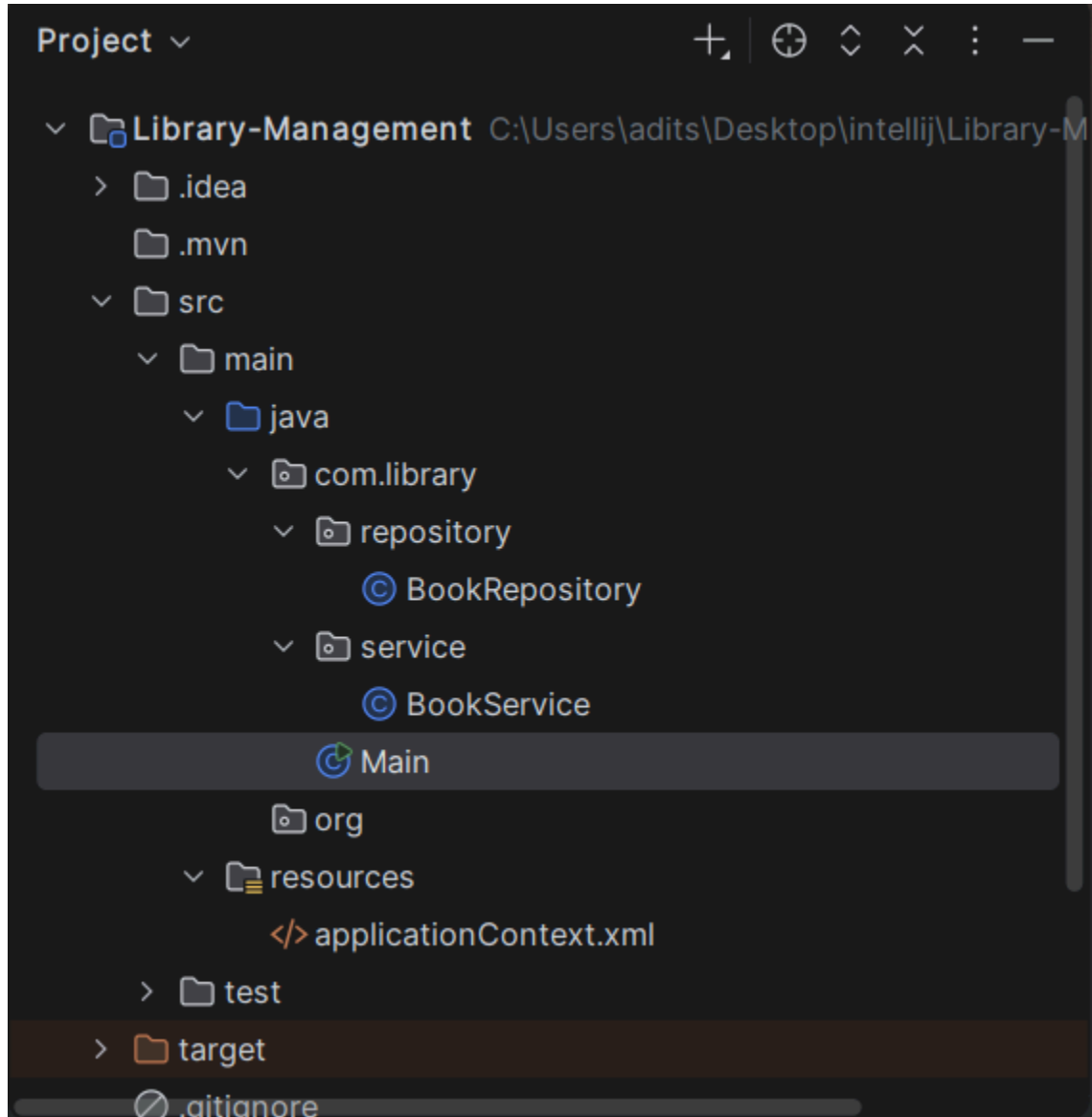


Exercise 2: Implementing Dependency Injection

File Structure:



Code:

BookRepository.java:

```
package com.library.repository;

public class BookRepository {

    public void showRepository() {
        System.out.println("Inside Book Repository");
    }
}
```

```
}  
}
```

BoorSerice.java

```
package com.library.service;  
  
import com.library.repository.BookRepository;  
  
public class BookService {  
  
    private BookRepository bookRepository;  
  
    // Setter Injection  
    public void setBookRepository(BookRepository bookRepository) {  
        this.bookRepository = bookRepository;  
    }  
  
    public void showService() {  
        System.out.println("Inside Book Service");  
        bookRepository.showRepository();  
    }  
}
```

Pom.xml:

```
<?xml version="1.0" encoding="UTF-8"?>  
<project xmlns="http://maven.apache.org/POM/4.0.0"  
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"  
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0  
http://maven.apache.org/xsd/maven-4.0.0.xsd">  
    <modelVersion>4.0.0</modelVersion>  
  
    <groupId>org.example</groupId>  
    <artifactId>Library-Management</artifactId>  
    <version>1.0-SNAPSHOT</version>  
  
    <properties>  
        <maven.compiler.source>25</maven.compiler.source>  
        <maven.compiler.target>25</maven.compiler.target>  
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>  
    </properties>  
  
    <dependencies>  
  
        <dependency>  
            <groupId>org.springframework</groupId>  
            <artifactId>spring-context</artifactId>  
            <version>5.3.30</version>
```

```
        </dependency>

    </dependencies>

</project>
```

ApplicationContext.xml:

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

       xsi:schemaLocation="
           http://www.springframework.org/schema/beans
           https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository"
          class="com.library.repository.BookRepository"/>

    <bean id="bookService"
          class="com.library.service.BookService">

        <property name="bookRepository"
                  ref="bookRepository"/>

    </bean>

</beans>
```

Main.java

```
package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Main {

    public static void main(String[] args) {

        ApplicationContext ctx =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService service =
            (BookService) ctx.getBean("bookService");
```

```
        service.showService();
    }
}
```

OUTPUT:

The screenshot displays the IntelliJ IDEA IDE interface. The left sidebar shows the project structure for 'Library-Management', with the 'Main' class selected under the 'com.library.service' package. The main editor window shows the code for 'Main.java', which is a public class with a main method. The main method initializes an 'ApplicationContext', retrieves a 'BookService' bean, and calls 'showService()' on it. It also creates a new 'BookRepository' instance and calls 'showRepository()' on it. The bottom panel shows the 'Run' output console, which displays the execution path: 'C:\Program Files\Java\jdk-25.0.2\bin\java.exe', followed by the output 'Inside Book Service' and 'Inside Book Repository', and finally 'Process finished with exit code 0'.

```
public class Main {
    public static void main(String[] args) {
        ApplicationContext ctx =
            new ClassPathXmlApplicationContext("applicationContext.xml");

        BookService bookService =
            (BookService) ctx.getBean("bookService");

        bookService.showService();
        BookRepository bookRepository = new BookRepository();
        bookRepository.showRepository();
    }
}
```

Run Main

"C:\Program Files\Java\jdk-25.0.2\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2025.3.3\lib\idea_rt.jar=57249" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8

Inside Book Service
Inside Book Repository

Process finished with exit code 0